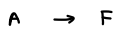
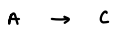
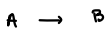


(1)



Stationary liquid

 C_A ?

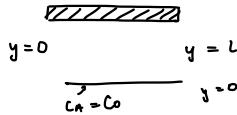
$C_A|_{y=0} = C_0$

$r_{AB} = -k_B$

$r_{AC} = -k_C$

$r_{AF} = -k_F C_A$

(20/50)



Balance Eq.

$$\frac{d}{dy} \left(D \frac{dC_A}{dy} \right) + k_B = 0$$

mass balance:

$$D \frac{d^2 C_A}{dy^2} - k_F C_A - k_B - k_C = 0$$

$$BC_1: C_A(y=L) = C_{AL}$$

$$-D \frac{dC_A}{dy} (y=0) = -k_1 C_A|_{y=0}$$

$$C_A(y) = -\frac{k_0}{2D} y^2 + Ay + B$$

$$BC_1 \Rightarrow C_{AL} = -\frac{k_0}{2D} L^2 + AL + B$$

$$BC_2 \Rightarrow -DA = -k_1 B \Rightarrow B = \frac{D}{k_1} A$$

$$\Rightarrow A = \frac{C_{AL} + \frac{k_0}{2D} L^2}{L + \frac{D}{k_1}} \quad \Rightarrow B = \frac{D \left[C_{AL} + \frac{k_0}{2D} L^2 \right]}{k_1 L + D}$$

$$A = \frac{k_1 \left[C_{AL} + \frac{k_0}{2D} L^2 \right]}{k_1 L + D}$$

$$C_A(x) = -\frac{k_0}{2D} x^2 + \frac{k_1 x + D}{k_1 L + D} \left[C_{AL} + \frac{k_0}{2D} L^2 \right]$$

70/100

The diagram shows a three-layer system. The top layer is labeled ϵ_L and has a thickness D_1 . The middle layer is labeled μ_1 and has a thickness D_2 . The bottom layer is labeled ϵ_R and has a thickness m_2 . The layers are separated by horizontal lines. The top boundary is at $x = -L$, the interface between the middle and bottom layers is at $x = 0$, and the bottom boundary is at $x = L$. A vertical line is drawn at $x = 0$ and another at $x = L$. The middle layer is shaded with diagonal lines.

$$(50/50)$$



$$C_{A_1} = A_1 + B$$

$$\frac{dC_{A2}}{dx} = \frac{K_0}{D_2} x + C_3$$

$$P_{C_2} : \phi_1 = \frac{C_{A_1}(x=0^-)}{C_{A_2}(x=0^+)}$$



$$\Rightarrow A = \frac{B - C_L}{L}$$

$$E = C_R - \frac{K_0}{2D_2} L^2 - D L = \frac{B}{\phi}$$

$$BC \approx 3: \quad \phi_1 = \frac{B}{E}$$

$$\Rightarrow D = \frac{D_1 A}{D_L} = \frac{(B - C_L) D_1}{L D_L}$$

$$C_R - \frac{k_0}{2D_2} L^2 - \frac{(B - CL)}{D_1} D_1 = \frac{B}{\phi}$$

$$C_R \phi D_L - \frac{k_0 \phi L^2}{2} - (B - C_L) D_1 \phi = B$$

$$B + (B - C_L) D_1 \phi = C_R \phi D_L - \frac{k_0 \phi L^2}{2}$$

$$B (1 + D_1 \phi) = C_R \phi D_L - \frac{k_0 \phi L^2}{2} + D_1 C_L \phi$$

$$\Rightarrow B = \frac{C_R \phi D_L - \frac{k_0 \phi L^2}{2} + D_1 C_L \phi}{1 + D_1 \phi}$$

$$\Rightarrow C_{A_1}(x) = \frac{C_R \phi D_L - \frac{k_0 \phi L^2}{2} - C_L}{(1 + D_1 \phi) L} x + \frac{C_R \phi D_L - \frac{k_0 \phi L^2}{2} + D_1 C_L \phi}{(1 + D_1 \phi)}$$

$$\Rightarrow C_{A_2}(x) = \frac{k_0}{2D_2} x^2 + \frac{C_R \phi D_L - \frac{k_0 \phi L^2}{2} - C_L}{(1 + D_1 \phi) L} \frac{D_1}{D_L} x + \frac{C_R - \frac{k_0}{2D_2} L^2 - D_1}{D_L}$$

$$N_{A_1}''(x) = -D_1 \frac{dC_{A_1}}{dx} = - \frac{(C_R \phi D_L - \frac{k_0 \phi L^2}{2} - C_L) D_1}{(1 + D_1 \phi) L}$$

$$N_{A_2}''(x) = -D_2 \frac{dC_{A_2}}{dx} = -D_2 \left(\frac{k_0}{2D_2} x + \frac{C_R \phi D_L - \frac{k_0 \phi L^2}{2} - C_L}{(1 + D_1 \phi) L} \frac{D_1}{D_L} \right)$$